

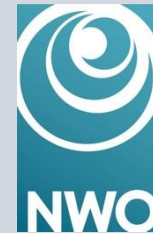
Bayesian Graphical Modeling

Theory

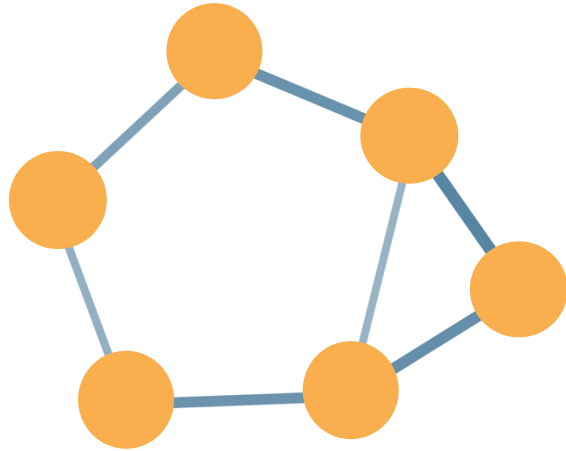
PsyNets Winterschool 2026

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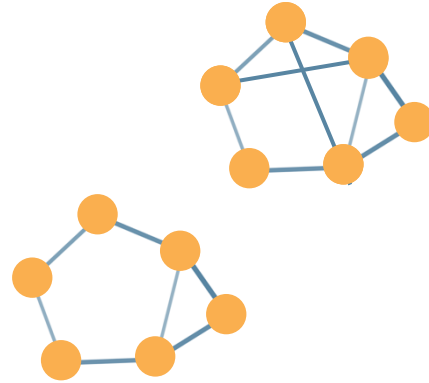


Setup Today



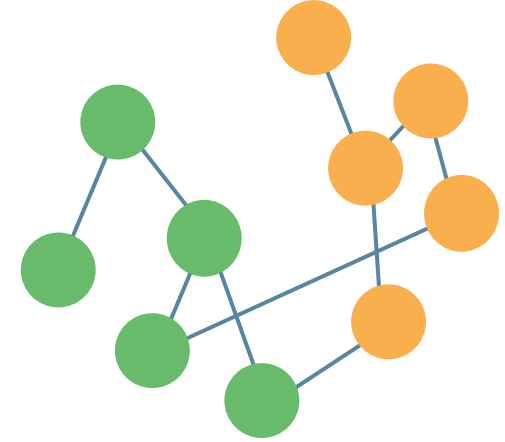
Estimating & Testing

Morning Session



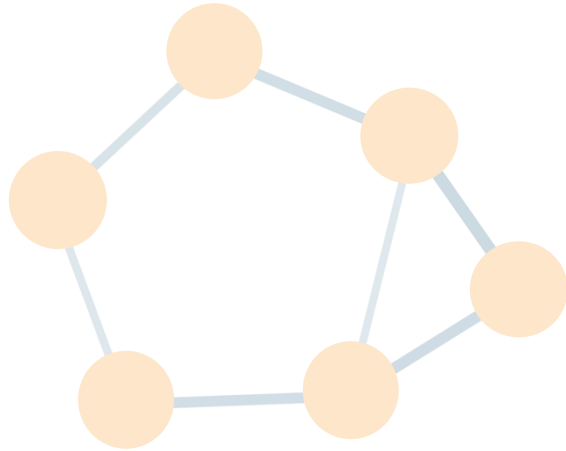
Network Comparison

Afternoon Session



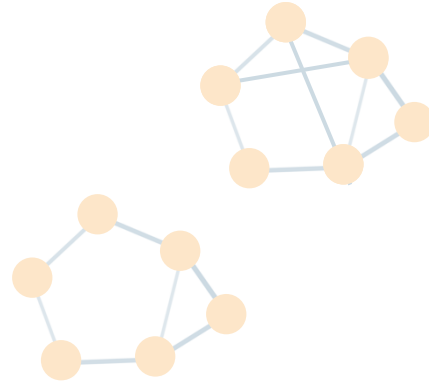
Clustering

Setup Today



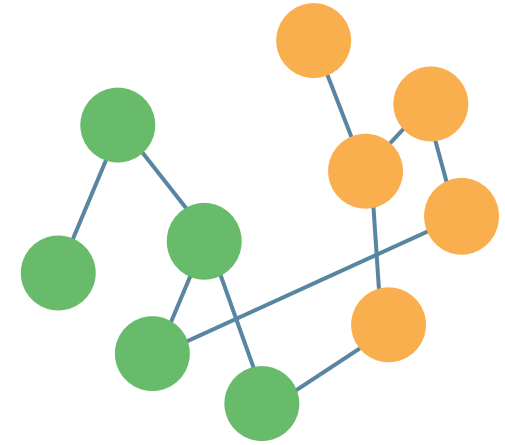
Estimating & Testing

Morning Session



Network Comparison

Afternoon Session



Clustering

Session setup



Part I: Theory



Part II: Practical



Part III: Project

Priors on the network structure

The prior on the structure, $p(\mathbf{G})$, says which network structures we consider plausible before seeing the data.

Specifying it is one of the defining features of Bayesian graphical modeling.

Two structure priors so far

Bernoulli — each edge is included with a fixed and independent probability.

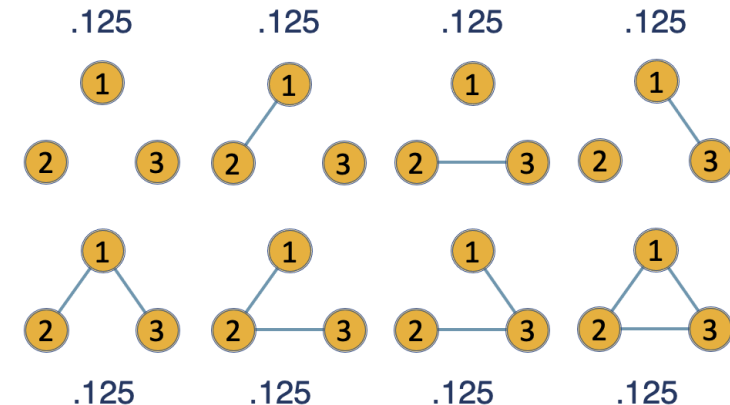
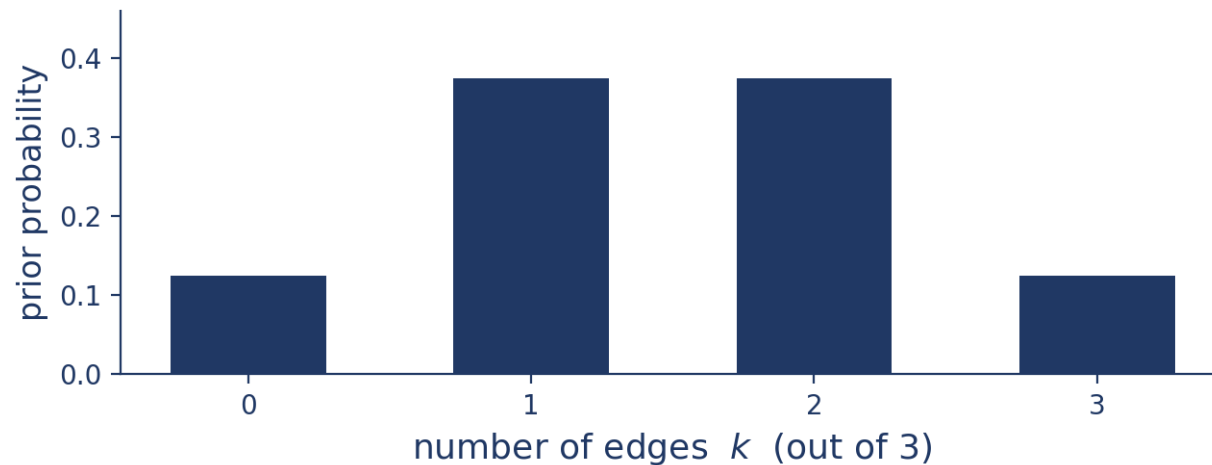
Beta–Bernoulli — a prior on the overall network density.

The Bernoulli prior

Each edge is included independently with a fixed probability π (default $\pi = 0.5$).

With $\pi = 0.5$ every network structure is equally likely a priori.

But the **density** is then binomial: most prior mass sits near half the edges, so very sparse or very dense networks are unlikely.

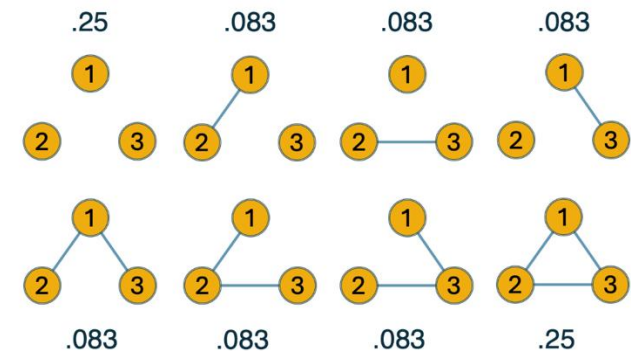
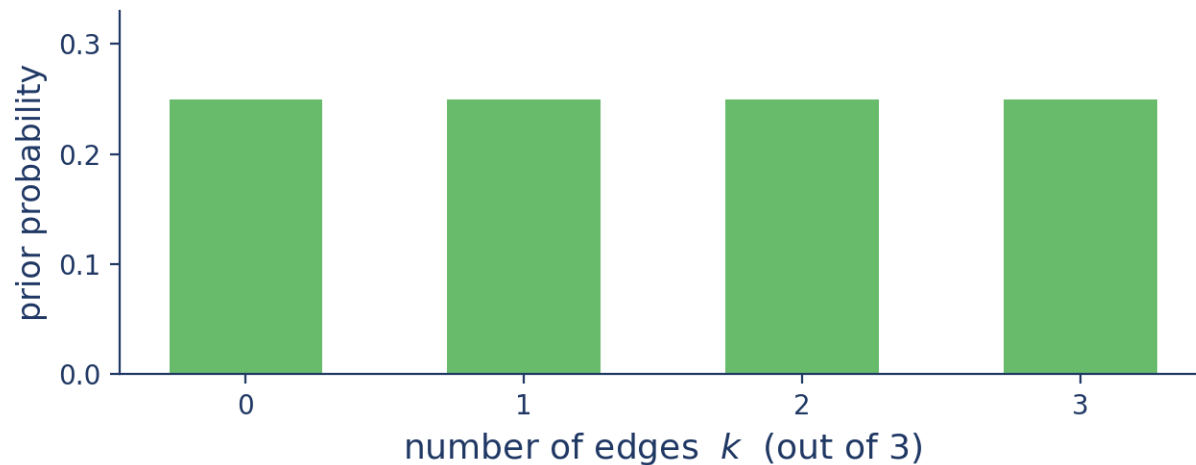


The Beta–Bernoulli prior

Instead of fixing π , place a **Beta(α , β)** prior on it

Beta(1, 1) makes every density equally likely (uniform over the number of edges).

This automatically **corrects for multiple testing**.



What they cannot express

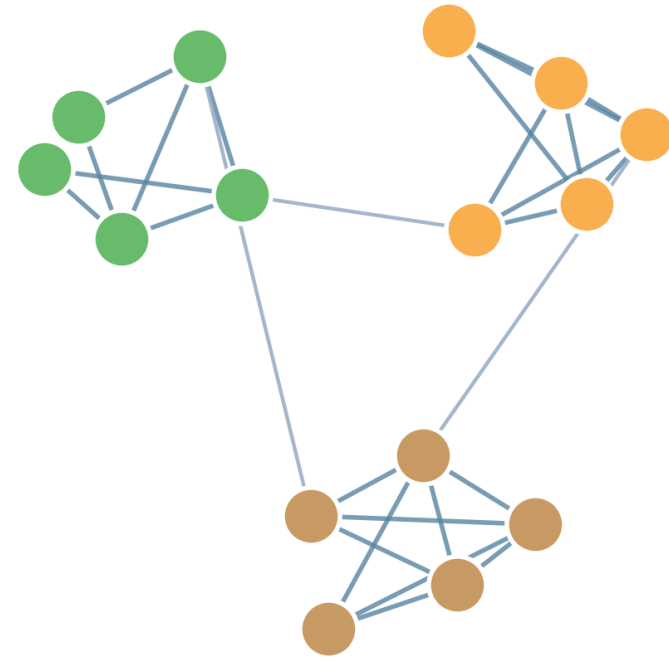
Neither prior can say that the variables form **clusters**.

Yet clustering is exactly what many psychological theories predict.

Clustering in psychological networks

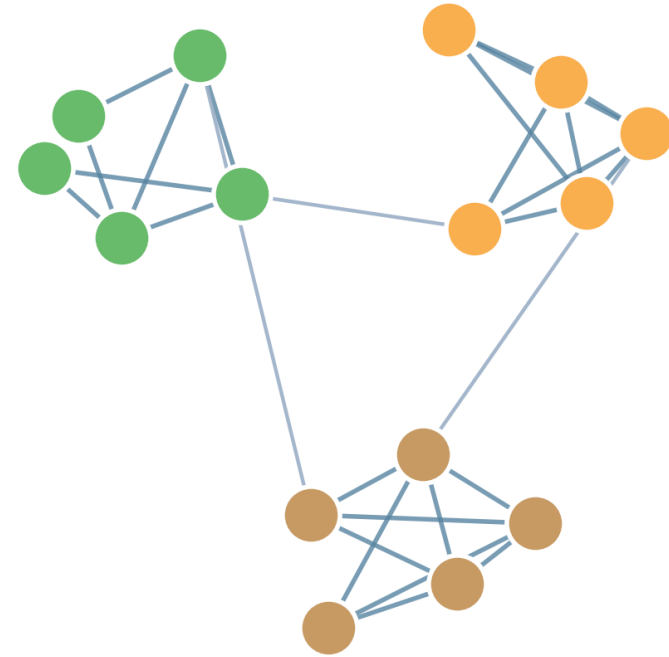
Intelligence, personality, and psychopathology all assume variables group into **clusters** (communities).

Connections are **dense within** a cluster and **sparse between** clusters — sometimes linked by a few bridge nodes.



What is a cluster?

A **cluster** (block / community) is a set of nodes that are more densely connected to each other than to the rest of the network.



Two ways to find clusters

Post-hoc: first estimate the network, then run a community-detection algorithm on it (e.g. EGA).

Model-based: build the clustering assumption into the prior itself and then estimate and test it.

The Stochastic Block Model takes the second route.

The Stochastic Block Model (SBM) prior

Put the clustering assumption directly into $\mathbf{p}(\mathbf{G})$.

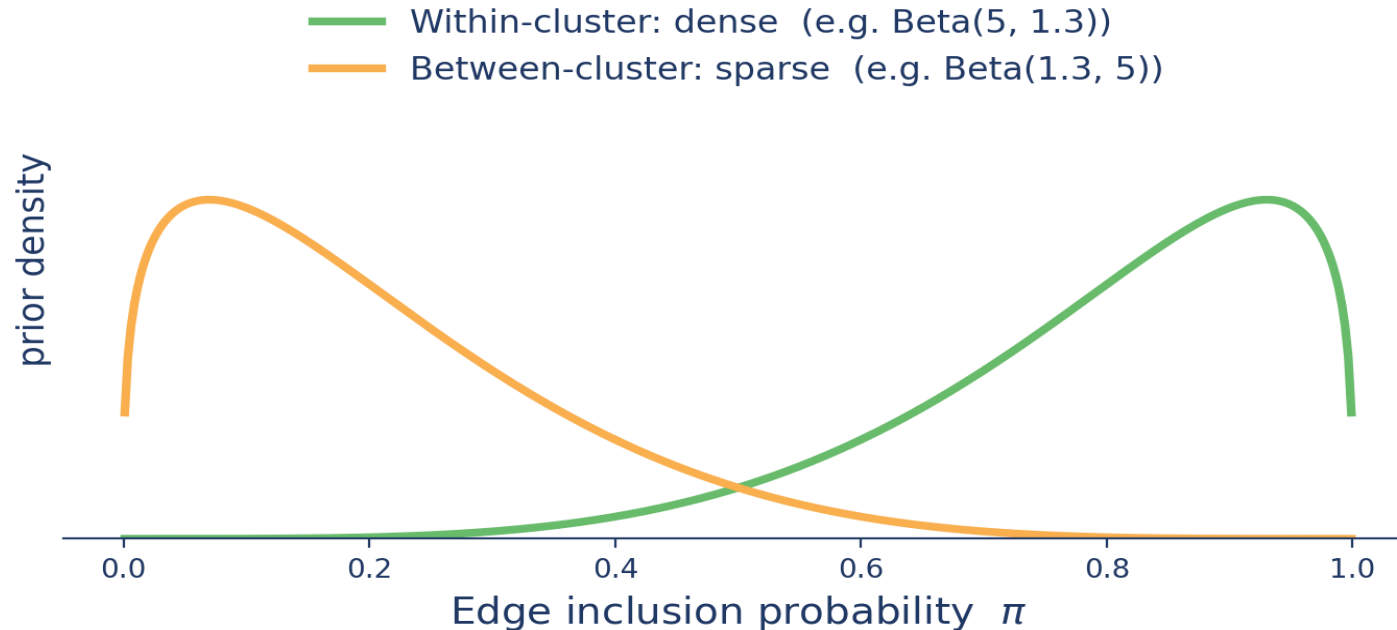
Each node belongs to one of **B latent blocks**; the label of node i is z_i .

The probability of an edge depends only on the blocks of the two nodes:

$$P(\text{edge between } i, j) = \pi_{z_i z_j}$$

Dense within, sparse between

Two separate **Beta–Bernoulli** priors on edge inclusion:
within a block — edges are more likely;
between blocks — edges are less likely.



The SBM **generalizes** the Beta-Bernoulli

With a single block (**$B = 1$**), every edge is within-cluster, so the SBM prior is exactly the Beta–Bernoulli prior.

With $B > 1$ it adds the possibility of multiple clusters

$$B = 1 \implies \text{SBM prior} = \text{Beta–Bernoulli prior}$$

The number of clusters B

Usually we do not know the number of clusters in advance.

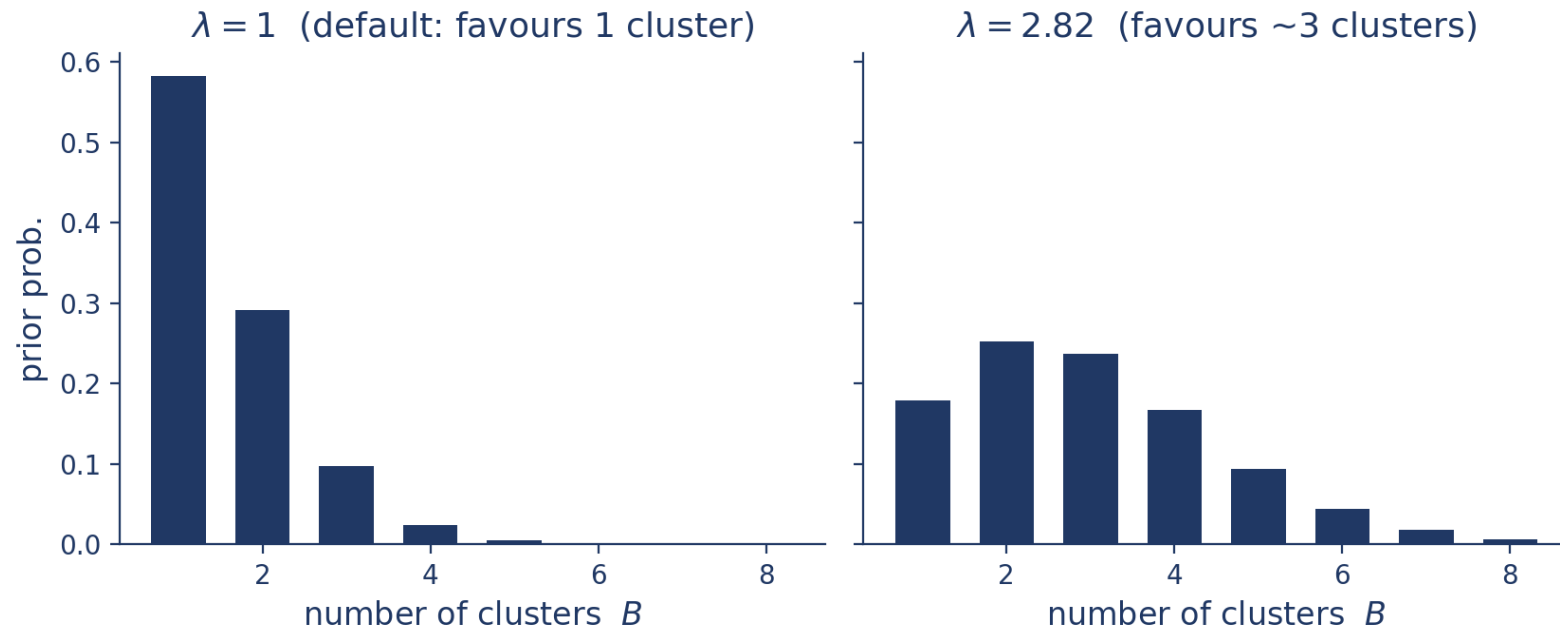
So the block labels $\mathbf{z}$ and the number of blocks $\mathbf{B}$ become parameters, estimated from the data.

We place a prior on B (a truncated Poisson):

$$B \sim \text{Poisson}^+(\lambda), \quad B \geq 1$$

Hyperparameter 1: how many clusters?

λ is the rate of the truncated Poisson on B : it controls the expected number of clusters a priori.



Hyperparameter 1: how many clusters?

λ is not exactly the expected number of clusters, but tracks it closely:

$\lambda = 1 \rightarrow \sim 1$ cluster (default)

$\lambda = 1.59 \rightarrow \sim 2$ clusters

$\lambda = 2.82 \rightarrow \sim 3$ clusters

$\lambda = 3.92 \rightarrow \sim 4$ clusters

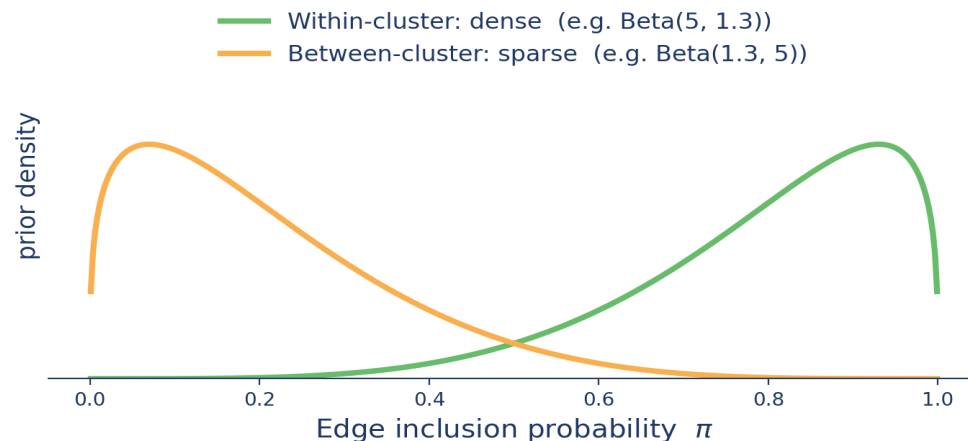
For 5 or more, set λ to the expected number.

Hyperparameter 2: within / between density

A Beta prior for **within-block** edges and a separate Beta prior for **between-block** edges.

Make within-block inclusion lean toward 1 and between-block lean toward 0 to encode dense-within, sparse-between.

Defaults are uniform, Beta(1, 1), for both.



Hyperparameter 3: cluster-size balance

A **Dirichlet(ϵ)** prior controls how evenly nodes spread across clusters.

Large $\epsilon \rightarrow$ similar-sized clusters; small $\epsilon \rightarrow$ some large, some tiny.

Default $\epsilon = 1$ is neutral

From prior to clustering analysis

With the SBM prior, clustering emerges during estimation. We can now:

- estimate the **number of clusters**,
- test** whether there is clustering at all,
- estimate **which nodes** cluster together.

Output 1: posterior over the number of clusters

How many clusters does the data support? Here, most posterior mass is on **$B = 2$** .



Output 2: testing for clustering

The **clustering Bayes factor** compares two hypotheses:

$$\mathcal{H}_1 : B > 1 \text{ (clustering)} \quad \text{vs.} \quad \mathcal{H}_0 : B = 1 \text{ (no clustering)}$$

$$\text{BF}_{10} = \frac{p(\mathcal{H}_1 | X) / p(\mathcal{H}_0 | X)}{p(\mathcal{H}_1) / p(\mathcal{H}_0)}$$

Output 2: testing for clustering

BF > 1 → evidence for clustering; **BF** < 1 → evidence against.

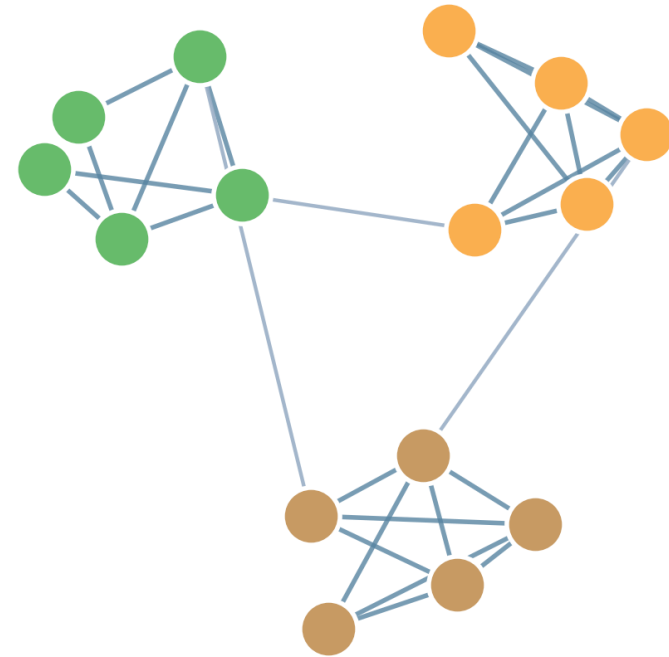
We can also compare two specific counts, e.g. **B = 2 vs B = 3** (a point hypotheses Bayes factor).

Useful when theory predicts a particular number of clusters.

Output 3: which nodes cluster together?

Each MCMC iteration assigns nodes to blocks.

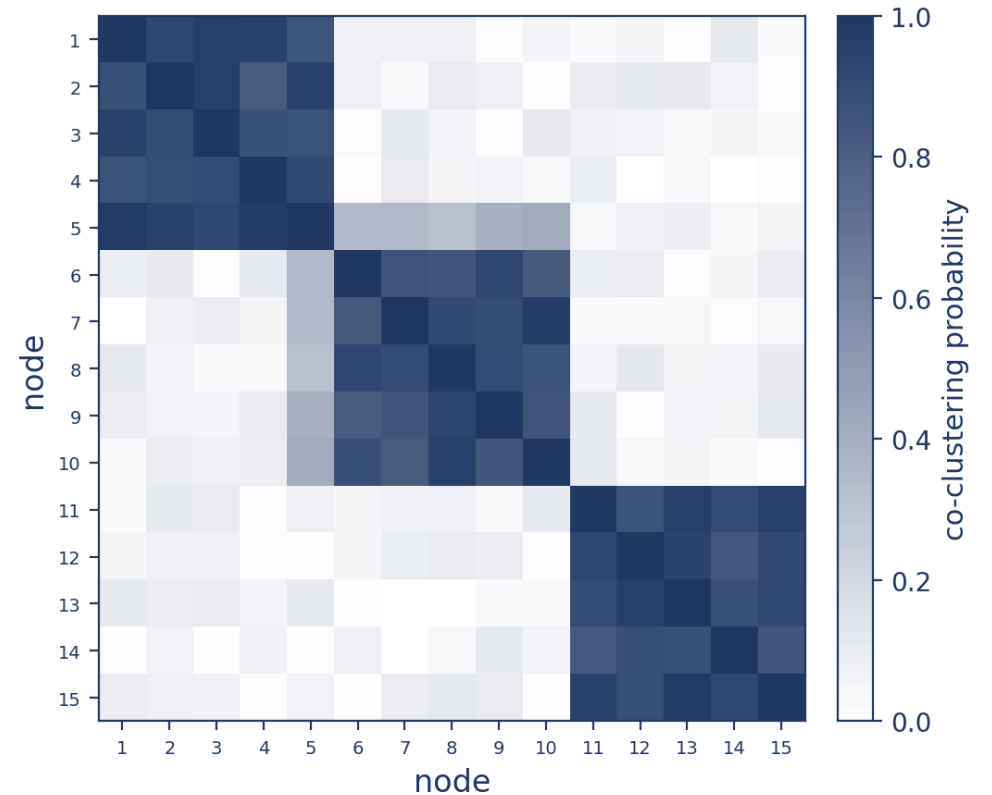
We summarize with the posterior **mode** and posterior **mean** allocation.



Output 3: the co-clustering matrix

For each pair of nodes: the proportion of iterations they shared a cluster.

Clear blocks = well-supported clusters; diffuse rows = **unstable** nodes (e.g. node 5).



Take-home: clustering as a testable hypothesis

The SBM prior turns a substantive theory (variables form clusters) into part of the statistical model.

We can **estimate** the clusters and **test** whether they exist, not just detect them post-hoc.

Bonus: where do the hyperparameters come from?

Choosing λ and the Beta shapes can be hard.

Recent work explores using **large language models** to elicit these priors from a description of the items and theory.

See: Sekulovski, Waaijers & Arena (2025), LLM-Based Prior Elicitation for Bayesian Graphical Modeling.

Session setup



Part I: Theory



Part II: Practical



Part III: Project